Laasya M. Koduru

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Education

University of California, Santa Barbara — Santa Barbara, CA
Doctor of Philosophy (Ph.D.), Computer Science — Sept 2024–Present
Advisors: Dr. Elizabeth Belding, Dr. Arpit Gupta

University of California, Santa Barbara — Santa Barbara, CA
Master of Science (M.S.), Computer Science — Sept 2023–Present

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science (B.S.), Applied Mathematics — Sept 2020–June 2023

Research Interests

Reliable and cost-efficient LLM and computer-use agents for large-scale web measurement, applied to data-driven broadband policy.

Selected Publications

[1] **Laasya Koduru**, Kira Allmann, Tejas Narechania, Arpit Gupta, "From Measurement to Policy Design: A Convex Optimization Framework for Broadband Affordability", Research Conference on Communications, Information and Internet Policy (TPRC), 2026.

[2] **Laasya Koduru**, Roman Beltiukov, Alexander Nguyen, Eugene Vuong, Jaber Daneshamooz, Tejas Narechania, Elizabeth Belding, Arpit Gupta, "Robust and Extensible Measurement of Broadband Plans with BQT+", arXiv preprint, Feb 2026.

[3] **Laasya Koduru**, Roman Beltiukov, Jaber Daneshamooz, Eugene Vuong, Arpit Gupta, Elizabeth Belding, Tejas Narechania, "Enabling Data-Driven Policymaking Using BQT+", PRIME Workshop at ACM IMC 2025.

[4] **Laasya Koduru**, Tejas Narechania, Elizabeth Belding, Arpit Gupta, "Poster: Enabling Data-Driven Policymaking Using the Broadband-Plan Querying Tool (BQT+)", Proceedings of the ACM IMC 2025 Conference.

[5] Jaber Daneshamooz, Eugene Vuong, **Laasya Koduru**, Sanjay Chandrasekaran, Arpit Gupta, "NetGent: Agent-Based Automation of Network Application Workflows", Workshop on Machine Learning for Systems at NeurIPS, Dec 2025.

[6] **Laasya Koduru**, Alejandro Alvarado Rojas, Angel Chavez Penate, Siyi Zhou, François Bar, Elizabeth Belding, Hernan Galperin, Arpit Gupta, "Assessing the Broadband Service Gaps and Affordability Barriers in BEAD-Eligible Areas", Research Conference on Communications, Information and Internet Policy (TPRC), 2025.

[7] **Laasya Koduru**, Arpit Gupta, Elizabeth Belding, Tejas Narechania, "Evaluating the Effects of and Interdependencies Among Federal Broadband Funding Programs", Research Conference on Communications, Information and Internet Policy (TPRC), 2025.

[8] Zhuowei Wen, Jocelyn Bliton, **Laasya Koduru**, Arpit Gupta, Shaddi Hasan, "Strategic Misreporting in the National Broadband Map", Research Conference on Communications, Information and Internet Policy (TPRC), 2025.

[9] Haarika Manda, Varshika Srinivasavaradhan, **Laasya Koduru**, Kevin Zhang, Xuanhe Zhou, Udit Paul, Elizabeth Belding, Arpit Gupta, Tejas Narechania, "The Efficacy of the Connect America Fund in Addressing US Internet Access Inequities", Proceedings of the ACM SIGCOMM 2024 Conference.

Full list at lmkoduru.github.io.

Policy Impact

- **Congressional Brief:** Provided the BQT+ data and analysis behind a briefing to congressional staff on Capitol Hill by my advisor, a UC Presidential Faculty Fellow.
- **Virginia Joint Commission on Technology and Science (JCOTS):** BQT+ data informed the affordability benchmarks in "Beyond Access: Broadband Affordability & Adoption" (co-author), Nov 2025.
- **New York City Office of Technology and Innovation (OTI):** BQT+ address-level availability and affordability analysis informed the City's Internet Access and Adoption Plan and its fiscal-impact assessment (co-author).
- **California Public Advocates Office (CalAdvocates), CPUC:** Contributed BQT pricing data to "Broadband Competition and Pricing Strategies in California's Urban Markets" (2026), whose analysis found over $1B per year in potential consumer savings.

Invited Talks & Tutorials

- "Informing Data-Driven Broadband Policy: Tools for Policymakers, Scholars, and Advocates," TPRC54 Pre-Conference Tutorial, American University Washington College of Law, Washington, DC — Sept 2026
- "Enabling Data-Driven Broadband Policymaking using BQT+," AIMS-19 Workshop, UC San Diego — Feb 2026
- "Data-Driven Broadband Policymaking: Bridging the Data Gaps for Effective Policymaking with BQT," Quello Center, Michigan State University — Dec 2025

Honors

- UC Santa Barbara Neal Fenzi Resonant Founder Fellowship — April 2026–Present
- The National GEM Consortium Ph.D. Science and Engineering Fellowship — April 2025–Present
- IETF/IRTF Applied Networking Research Prize (SIGCOMM 2024 Connect America Fund paper) — April 2025
- UC Santa Barbara Computer Science Summer Fellowship — June 2024

Experience

University of California, Santa Barbara — Santa Barbara, CA
Graduate Student Researcher — Sept 2023–Present
Systems & Networking Lab (SNL) and Mobility Management & Networking Lab (MOMENT)

- Designed BQT+, an agentic system in which frontier LLMs discover each ISP's address-checker workflow from a natural-language goal and emit a reusable workflow specification. Covers 100+ ISPs.
- Replays cached specifications across addresses at scale, with 100% agreement on 500 address–ISP pairs. Measured 56 website changes across 64 ISPs in eight months.
- Released public broadband plan data and a natural-language query interface used by Virginia JCOTS, NYC OTI, and CalAdvocates.
- Deployed BQT+ on Kubernetes in Google Cloud with autoscaling and load balancing across servers.

Idaho National Laboratory — Idaho Falls, ID
Cybersecurity Machine Learning Algorithms Intern — June–Sept 2025

- Built a bidirectional LSTM model using PyTorch and TensorFlow that separates mixed Ethernet communication signals into their original source signals, reaching 96% accuracy against a 90% baseline.
- Ran a grid search over 200+ hyperparameter configurations on INL's Hoodoo NVIDIA A100 GPU cluster, then trained the final model on the cluster.
- Enabled new capabilities for securing control systems in critical infrastructure. Patent application under review.

Projects

CVE-Genie: Automated CVE Reproduction | *Python* Sept–Dec 2024

- Co-built an LLM multi-agent system that rebuilds, exploits, and verifies CVEs from public reports.
- Reproduced 50 CVEs end to end, including C/C++ memory-safety bugs.

MOVERC: MongoDB Version Classifier | *Python* April–June 2024

- Built a random forest that infers MongoDB versions (3.0–7.0) from packet headers to flag vulnerable servers.
- Generated a 30K-flow dataset from 220 MongoDB apps under varied network conditions (netMosaic, NTLFlowLyzer).
- Reached 0.79 precision and 0.74 weighted F1 against 0.53 and 0.42 for a neural net baseline. Audited the model through Trustee and ablation tests.

CodeInsight | *Java* March 2024

- Built an LLM tool on GPT and the VS Code Java debugger that answers "why" questions about program output.

Technical Skills

- **Languages:** Python, C, C++, Java
- **ML/AI:** PyTorch, TensorFlow, LLM agents
- **Data and Cloud:** Apache Spark, Kubernetes, Docker, Google Cloud, AWS
- **Tools:** Git, SQL, Selenium

Leadership & Service

UCSB CS Grad Student Association (GSA)

Vice President, Secretary Sept 2024–Oct 2026

- Represent CS graduate students at campus-wide GSA meetings; serve on the UCSB AI committee, speak at open houses, plan events.

Early Research Student Program, UC Santa Barbara

Ph.D. Mentor Sept 2025–June 2026

- Mentor a team of undergraduate students through an end-to-end research project, guiding research methodology and technical execution.

ACM Internet Measurement Conference (IMC) 2026

Shadow Technical Program Committee Member

- Review paper submissions for acceptance to IMC 2026.